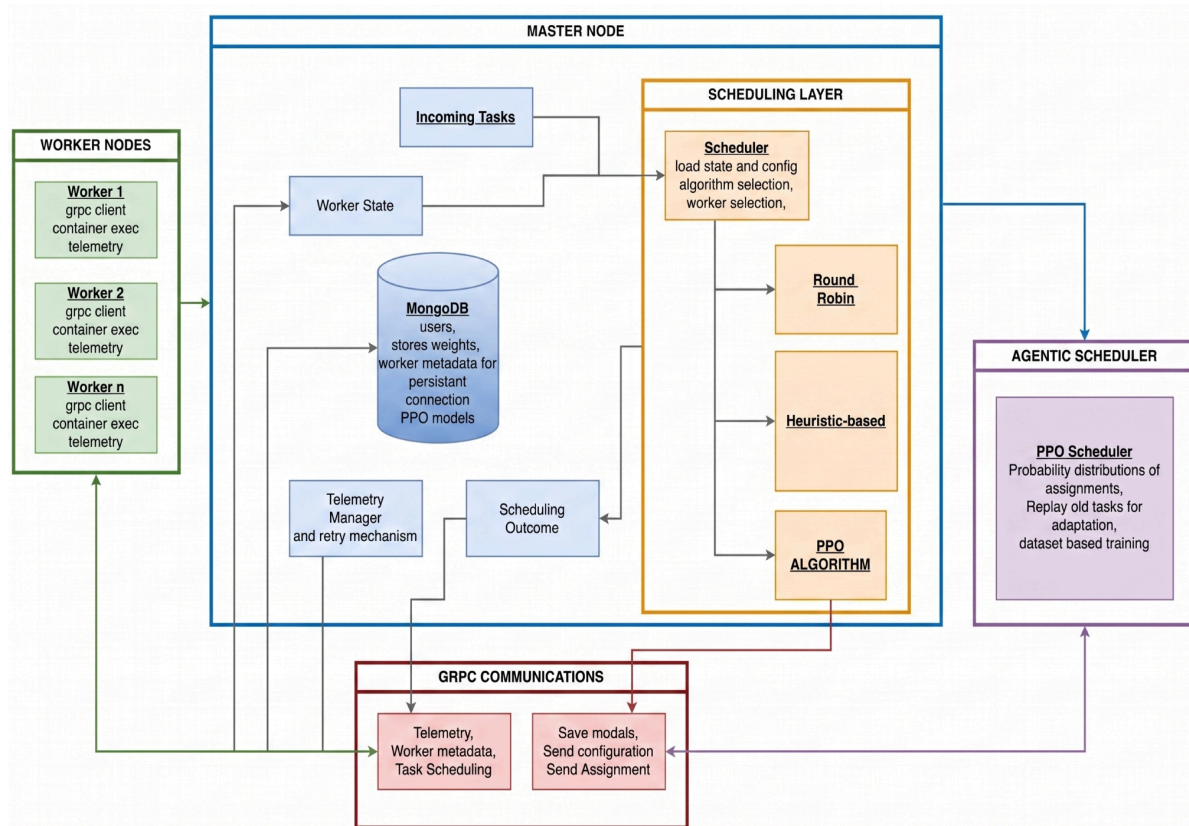


Agentic Cloud Cluster

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Modern cloud datacenters face scheduling challenges when assigning containerized tasks to heterogeneous worker nodes with varying resource capacities. Traditional approaches like Round-Robin and First-Fit fail to optimize latency, SLA compliance, often causing inefficient resource utilization and higher tail latency. This project addresses these limitations by developing an intelligent scheduling framework that adapts dynamically to system conditions.



The proposed solution uses Proximal Policy Optimization (PPO), a reinforcement learning algorithm that learns optimal scheduling policies from real-world workloads. The system includes a Go-based master node for orchestration, Docker-based worker nodes to simulate heterogeneity, MongoDB for persistence, and a React dashboard for monitoring. The PPO scheduler runs as a Python gRPC service, enabling low-latency decision making. The scheduling problem is modeled as a Markov Decision Process, where the agent observes task requirements and worker states through a 14-dimensional state space. A neural network-based policy outputs placement decisions, trained using Generalized Advantage Estimation with a reward function balancing latency, queue pressure, and load distribution. Training uses ~200,000 tasks from Alibaba's cluster trace dataset. Experimental results show improvements over baseline schedulers, including reduced makespan, lower turnaround time, and improved P95 latency. The system also includes benchmarking, trace replay, and observability features, demonstrating that reinforcement learning-based scheduling can outperform traditional methods.